

Semester: **Fall 2022**
Course Code: **CSE460**
Course Title: **VLSI Design**

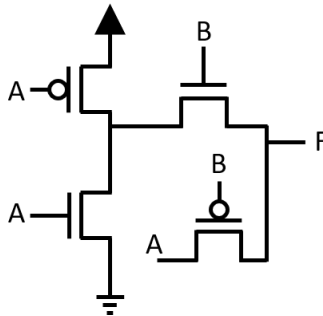
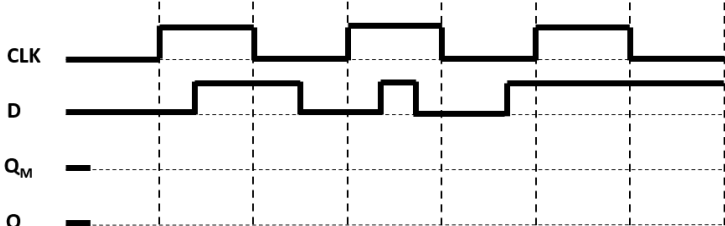
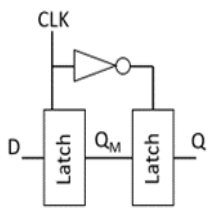
Midterm Exam
Full Marks: **15 x 3 = 45**
Time: **1 hour 30 minutes**
Date: **11th November 2022**

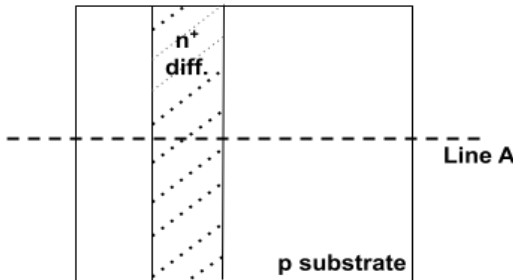
Set A

Student ID:	Name:	Section:
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[Answer any **THREE** questions out of **FOUR**. Each question carries equal marks.]

[After the exam, the question paper should be turned in along with the answer script.]

1. (CO2)	(a)	Complete the truth table shown in Figure 2, for the circuit shown in Figure 1. <table><thead><tr><th>A</th><th>B</th><th>F</th></tr></thead><tbody><tr><td>0</td><td>0</td><td></td></tr><tr><td>0</td><td>1</td><td></td></tr><tr><td>1</td><td>0</td><td></td></tr><tr><td>1</td><td>1</td><td></td></tr></tbody></table> Figure 1: A logic circuit Figure 2: Truth table	A	B	F	0	0		0	1		1	0		1	1		4
A	B	F																
0	0																	
0	1																	
1	0																	
1	1																	
	(b)	A D flip-flop is made by cascading two D latches as shown in Figure 3. Data input (D) and clock (CLK) for the circuit is shown in Figure 4. Draw the waveforms for Q_M and Q. (<u>Use a pencil to draw the signals and draw on the question paper, no need to draw the diagram in your answer script separately.</u>)  Figure 3 Figure 4	10															
	(c)	Identify the type of the flip-flop shown in figure 3.	1															

2. (CO1)	Observe figure 5 and answer the following questions.		
	 Figure 5: Top view of a wafer after fabricating an n+ diffusion region (dotted region)		
	(a)	Draw the cross-sectional view of the above figure along ‘Line A’ and label the different regions.	3
(b)	Briefly describe and illustrate the necessary steps to fabricate the n+ diffusion region on a blank wafer as shown in the figure.		12
3. (CO2)	(a)	Design a CMOS compound gate that implements the following function: $Y = (A+B).(C+D) + E$	6
	(b)	Imagine a <i>gate-level</i> logic circuit with 1 output f , and 4 inputs x1 , x0 , y1 , and y0 . - Let $X = x1x0$ be a two-digit binary number, where the four possible values of X are 00, 01, 10, and 11, which represent the four decimal numbers 0, 1, 2, and 3 respectively. - Let $Y = y1y0$ be another number with the same four possible values. - The output f should be 1 if the sum of numbers represented by <i>X</i> and <i>Y</i> is greater than decimal 3. Otherwise, the output f should be 0.	
		i) Create a truth table for all 16 combinations of the 4 inputs.	3
		ii) Derive the boolean expression of the output using k-map.	4
		iii) Implement the function f with logic gates.	2
4. (CO2)	The input-output behavior of a sequence detector FSM is given below.		
	CLK	t ₁ t ₂ t ₃ t ₄ t ₅ t ₆ t ₇ t ₈ t ₉ t ₁₀ t ₁₁ t ₁₂ t ₁₃ t ₁₄ t ₁₅ t ₁₆ t ₁₇ t ₁₈	
	w	1 0 0 1 1 1 0 0 0 0 0 0 0 0 1 0 0 0	
	z	0 1 0 1 0 0 1 0 0 0 0 0 0 0 1 1 0 0	
	(a)	Identify the <u>sequence</u> and the <u>type</u> of FSM for the above input-output combinations. [<u>Hint</u> : The above table contains two sequences]	1+1
(b)	Design the state diagram, the state table, and the gray-encoded state assigned table. [Overlapping allowed].		3+2+2
(c)	Determine the logic expressions of the next state and output variables using k-maps.		6